

LOW ACRYLONITRILE NBR PATENT REPORT: EVIDENCE FROM US11347156B2

1. ABSTRACT

This report synthesizes the patent evidence available in the required manifest for low acrylonitrile NBR, with the manifest containing a single primary patent: US11347156B2. The patent is not directed to NBR synthesis or monomer-feed control; instead, it concerns an electro-conductive member for electrophotography that uses a matrix-domain rubber architecture. Within that architecture, acrylonitrile-butadiene rubber (NBR) appears as one of the candidate diene-based rubbers for either the matrix or the domains, paired with styrene-butadiene rubber (SBR) in the complementary phase. The extracted evidence includes explicit nitrile-content values for several commercial NBR grades used as reference materials in the examples, ranging from 15% to 41.5%, which provides context for low- versus higher-acrylonitrile grades, but these values are not disclosed as properties of the claimed target material itself. The patent's core technical contribution is phase-separated rubber design: a first rubber composition forms the matrix, a second rubber composition forms conductive domains, and the two rubbers are selected to differ in monomer units and to have controlled solubility-parameter separation and viscoelastic matching. The examples emphasize formulation-driven performance rather than polymerization chemistry, using NBR, SBR, IR, BR, EPDM, CR, and hydrix as raw rubbers, along with carbon black, sulfur, and vulcanization accelerators. Water is not part of the polymerization process because no polymerization is disclosed; the evidence instead shows dry compounding, molding, and evaluation of electro-conductive members. Overall, the patent is only indirectly relevant to low acrylonitrile NBR because it documents NBR grades and their use in a functional rubber composition, but it does not disclose NBR preparation, acrylonitrile feed control, or synthesis conditions for low-acrylonitrile NBR.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US11347156B2
- Patent Title: Electro-conductive member for electrophotography, process cartridge for ...
- Assignee: Canon Kabushiki Kaisha

- Jurisdiction: US
- Publication Year: 2022
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent does not disclose NBR synthesis or polymerization; it discloses fabrication of an electro-conductive member by compounding rubber compositions into a matrix-domain structure and molding them onto an electro-conductive support. NBR is used as one of the rubber materials in the examples and claims, paired with SBR in the complementary phase. - Nitrile content: 15% — Example section, NBR (1) JSR NBR N260S - SP value: 17.2 (J/cm³)^{0.5} — Example section, NBR (1) JSR NBR N260S - Nitrile content: 41.5% — Example section, NBR (2) JSR NBR N220S - SP value: 20.6 (J/cm³)^{0.5} — Example section, NBR (2) JSR NBR N220S - Nitrile content: 27.5% — Example section, NBR (3) Nipol DN302 - SP value: 18.8 (J/cm³)^{0.5} — Example section, NBR (3) Nipol DN302 - Nitrile content: 18.0% — Example section, NBR (4) Nipol DN401LL - SP value: 17.4 (J/cm³)^{0.5} — Example section, NBR (4) Nipol DN401LL - Nitrile content: 35% — Example section, NBR (5) Nipol N230S - SP value: 20.0 (J/cm³)^{0.5} — Example section, NBR (5) Nipol N230S - Nitrile content: 40.0% — Example section, NBR (6) JSR NBR N202S - SP value: 20.4 (J/cm³)^{0.5} — Example section, NBR (6) JSR NBR N202S - Raw rubber: NBR 100 parts by mass — Example 1, Table 1 - Raw rubber: Unvulcanized domain composition 30 parts by mass — Example 1, Table 2 - Raw rubber: Polyisoprene 70 parts by mass — Example 1, Table 2 - Filler: Calcium carbonate 40 parts by mass — Example 1, Table 2 - Vulcanization accelerating aid: Zinc oxide 5 parts by mass — Example 1, Table 2 - Processing aid: Zinc stearate 2 parts by mass — Example 1, Table 2 - Vulcanizing agent: Dispersive sulfur 3 parts by mass — Example 1, Table 3 - Vulcanization accelerator (1): Di-2-benzothiazolyl disulfide 2 parts by mass — Example 1, Table 3 - Vulcanization accelerator (2): Tetramethylthiuram disulfide 0.5 parts by mass — Example 1, Table 3 - Electro-conductive particles in domains: 50 to 150 parts by mass per 100 parts by mass of the second rubber — Claim 1 - Difference of absolute values of SP values: 0.2 to 4.0 (J/cm³)^{0.5} — Claim 1 - $\tan \delta_1 / \tan \delta_2$: 0.45 to 2.00 — Claim 1 - Temperature: 23° C. — claim/evaluation context - Relative humidity: 50% — claim/evaluation context - Frequency: 80 Hz — claim/evaluation context - Domain volume fraction: 28.7 vol% — Table 5-1, Example 1 - $\tan \delta_1 / \tan \delta_2$ (A): 0.57 — Table 5-1, Example 1 - $\tan \delta_1 / \tan \delta_2$ (B): 0.23 — Table 5-1, Example 1 - Domain volume fraction: 29.1 vol% — Table 5-1, Example 2 - $\tan \delta_1 / \tan \delta_2$ (A): 0.46 — Table 5-1, Example 2 - $\tan \delta_1 / \tan \delta_2$ (B): 0.17 — Table 5-1, Example 2 - Domain volume fraction: 28.9 vol% — Table 5-1, Example 3 - $\tan \delta_1 / \tan \delta_2$ (A): 0.50 — Table 5-1, Example 3 - $\tan \delta_1 / \tan \delta_2$ (B): 0.21 — Table 5-1, Example 3 - Domain volume fraction: 20.6 vol% — Table 5-1, Example 22 - $\tan \delta_1 / \tan \delta_2$ (A): 1.00 — Table 5-1, Example 22 - $\tan \delta_1 / \tan \delta_2$ (B): 1.38 — Table 5-1, Example 22 - Domain volume fraction: 30.6 vol% — Table 5-1, Example 26 - $\tan \delta_1 / \tan \delta_2$ (A): 0.67 — Table 5-1, Example 26 - $\tan \delta_1 / \tan \delta_2$ (B): 0.52 — Table 5-1, Example 26 - Domain volume fraction: 19.4 vol% — Table 5-1, Example 30 - $\tan \delta_1 / \tan \delta_2$ (A): 0.96 — Table 5-1, Example 30 - $\tan \delta_1 / \tan \delta_2$ (B): 1.87 — Table 5-1, Example 30 - Temperature: 23° C. — Example 40 evaluation - Relative humidity: 50% — Example 40 evaluation - Abutting pressure: 1250 gf (12.26 N) — Example 40 evaluation - Output speed: 370 mm/sec —

Example 40 evaluation - Image resolution: 1,200 dpi — Example 40 evaluation - Printing rate: 1.0% — Example 40 evaluation - Continuous output: 20000 sheets — Example 40 evaluation - Rest time: 12 hours — Example 40 evaluation - Conditioning time: 48 hours — Example 40 evaluation - Medium & Water Role: No polymerization medium is disclosed. Water is not used in synthesis, but the evaluation conditions explicitly include 23° C. and 50% relative humidity; no aqueous polymerization, coagulation, washing, or latex step is disclosed in the extracted evidence. - low acrylonitrile content: Not disclosed in extracted evidence

Medium & Water Role - No polymerization medium is disclosed. Water is not used in synthesis, but the evaluation conditions explicitly include 23° C. and 50% relative humidity; no aqueous polymerization, coagulation, washing, or latex step is disclosed in the extracted evidence. - Core reaction medium: No polymerization/reaction medium disclosed; the patent describes dry compounding/kneading of rubber compositions and molding of an electro-conductive member.

low acrylonitrile content - Not disclosed in extracted evidence

Relevant Experimental Evidence - Comparative Example 3: NBR was used as the matrix rubber with EPDM as the domain rubber, showing poor discharge-related performance due to large SP mismatch. - Comparative Example 5: NBR was used as the matrix rubber with SBR as the domain rubber, but the accelerator choice led to an unfavorable $\tan \delta_1/\tan \delta_2$ balance. - Comparative Example 6: NBR was used for both domain and matrix, and the matrix-domain structure could not be confirmed. - Example 1: NBR was used as the raw rubber in the domain composition, and the member was fabricated by kneading, molding, and evaluation. - Examples 2 to 39: multiple rubber pairings were compared, including NBR/SBR combinations, to correlate SP difference and viscoelasticity with image performance.

Technical Relevance This patent is relevant only as an application-side reference because it explicitly includes NBR grades and shows how NBR can be formulated in a phase-separated rubber structure. It does not disclose low-acrylonitrile NBR synthesis, but it provides composition context for NBR materials with nitrile contents as low as 15% in the example set.

PATENT COMPARISON TABLE

Patent	Applicant	Priority	Medium & Water Role	low acrylonitrile content	Key Findings
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US11347156B2	Canon Kabushiki Kaisha	2022	No polymerization medium is disclosed. Water is not used in synthesis, but the evaluation conditions explicitly include 23° C. and 50% relative humidity; no aqueous polymerization, coagulation, washing, or latex step is disclosed in the extracted evidence.	Not disclosed in extracted evidence	Comparative Example 3: NBR was used as the matrix rubber with EPDM as the domain rubber, showing poor discharge-related
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3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- Fewer than two selected primary patents were available, so cross-patent quantitative trends were not generated.

4. CONCLUSION

The single primary patent in the manifest, US11347156B2, is not a synthesis patent for low-acrylonitrile NBR; it is a formulation and device patent that uses NBR as one of several rubber materials in a matrix-domain electro-conductive member. The evidence provides useful compositional context, including explicit nitrile contents for several commercial NBR grades, but it

does not disclose preparation conditions, monomer feed ratios, or polymerization routes for low-acrylonitrile NBR.

5. REFERENCES

- US11347156B2 | Electro-conductive member for electrophotography, process cartridge for ... | Canon Kabushiki Kaisha | US | 2022 | <https://patents.google.com/patent/US11347156B2/en>